

Space Computing: Research & Engineering Practice on Data Management

Volume II: Technical Architecture & Analysis of Mainstream Solutions

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Acronyms

AHP	Analytic Hierarchy Process
AI	Artificial Intelligence
API	Application Programming Interface
CCSDS	Consultative Committee for Space Data Systems
CERN	European Organization for Nuclear Research
CFDP	CCSDS File Delivery Protocol
cFS	Core Flight System
COTS	Commercial Off-The-Shelf
DTN	Delay/Disruption Tolerant Networking
ESA	European Space Agency
FAIR	Findable, Accessible, Interoperable, and Reusable
FDIR	Fault Detection, Isolation, and Recovery
FPGA	Field-Programmable Gate Array
ISO	International Organization for Standardization
LEO	Low Earth Orbit
NASA	National Aeronautics and Space Administration
RTOS	Real-Time Operating System
SoM	System on Module
SWaP	Size, Weight, and Power
TRL	Technology Readiness Level
HEP	High-Energy Physics

Contents

Acronyms	2
1 Technical Architecture of Space Data Management	4
1.1 Five-Layer System Stack	4
1.2 Five-Module Functional Decomposition	4
1.3 Data Management Foundations & Standards Landscape	6
2 Panorama of Mainstream Technical Routes	6
2.1 Traditional Aerospace Dedicated Systems	6
2.2 Cloud-Native Edge Frameworks	6
2.3 High-Energy Physics Distributed Data Platforms	7
2.4 Commercial Aerospace Integrated Solutions	7
3 Comparative Analysis of Mainstream Solutions	7
3.1 Evaluation Framework and Methodology	7
3.2 Quantitative Scoring Results	8
3.3 Resource Consumption Comparison	9
4 Scenario Adaptability Analysis	9
5 Summary of This Volume	10
Technical Supplement	11
A1 Evaluation Methodology	11
A1.1 Weight Calibration	11
A1.2 Scoring Process	11
A2 Resource Consumption Measurement	12
A2.1 Baseline Hardware Configuration	12
A2.2 Measurement Protocol	12
A3 Scenario Adaptability Assessment	13
A3.1 Scoring Criteria	13
References	14

1 Technical Architecture of Space Data Management

1.1 Five-Layer System Stack

The unified technical architecture of modern space computing data management is fundamentally structured as a five-layer system stack. Progressing vertically from the underlying hardware interfaces to the highest-order application enablement planes, this taxonomy imposes a definitive division of responsibilities while mandating standardized interoperability across layer boundaries (see Figure 1):

1. **Hardware Abstraction Layer:** Provides a unified software abstraction over strictly heterogeneous on-board hardware — including System on Module (SoM), Field-Programmable Gate Array (FPGA), radiation-tolerant storage module, and avionics bus interface — effectively shielding upper-tier software functions from arbitrary hardware variances;
2. **Core Data Management Layer:** Executes mission-critical data persistence, algorithmic retrieval, and robust transmission routing. This incorporates distributed storage schema, dynamic replica management, checksum-based integrity protection, and stateful space-ground data synchronization;
3. **Resource Orchestration Layer:** Autonomously governs and orchestrates the finite reservoir of on-board Size, Weight, and Power (SWaP) variables. Tasks encompass heterogeneous compute pipeline scheduling, real-time dynamic memory allocation, and power-aware workload throttling;
4. **Fault Tolerance & Reliability Layer:** Enforces space-environment resilience through error detection and correction, multi-tiered Fault Detection, Isolation, and Recovery (FDIR), and the maintenance of deterministic states critical for long-duration autonomous operability;
5. **Application Interface Layer:** Exposes secure, standardized Application Programming Interfaces (APIs) to upper-layer mission applications, structurally supporting the rapid integration, continuous deployment, and on-orbit reconfiguration of intelligent space payloads.

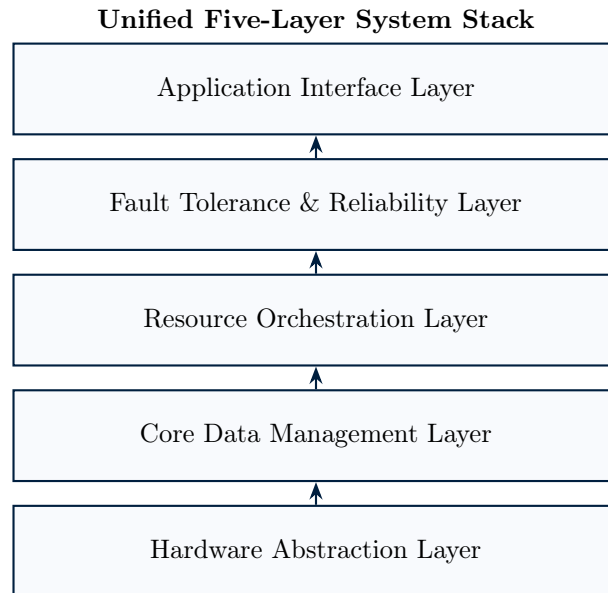


Figure 1: Unified Five-Layer Technical Architecture Stack

1.2 Five-Module Functional Decomposition

Zooming conceptually into the Core Data Management Layer, the architecture can be logically distilled into five highly interdependent functional modules (see Figure 2). These modules interoperate synchronously to guarantee end-to-end data pipeline integrity:

1. **Data Ingestion Module:** Curates the high-throughput reception and immediate preprocessing of raw sensor streams, conducting localized data validation, standardized format translation, and initial zero-value filtering;
2. **Storage & Indexing Module:** Facilitates highly deterministic on-board state persistence, supporting distinct schemas for both structured telemetry and unstructured payload matrices, equipped with topology-aware replication engines for constellation continuity;
3. **Processing & Analytics Module:** Furnishes dedicated on-board computation logic encompassing lossless compression, spatial feature extraction, and seamless integration pipelining with on-board edge Artificial Intelligence (AI) inferencing cores;
4. **Transmission & Scheduling Module:** Dictates space-ground and inter-satellite network transmissions. It natively implements link-aware rate adaptation, dynamic priority queuing, and fully stateful protocol enforcement;
5. **Monitoring & Control Module:** Operates as the autonomic nervous system of the software stack, generating comprehensive health diagnostics, granular resource consumption telemetry, and facilitating secure on-orbit workflow reconfigurations.

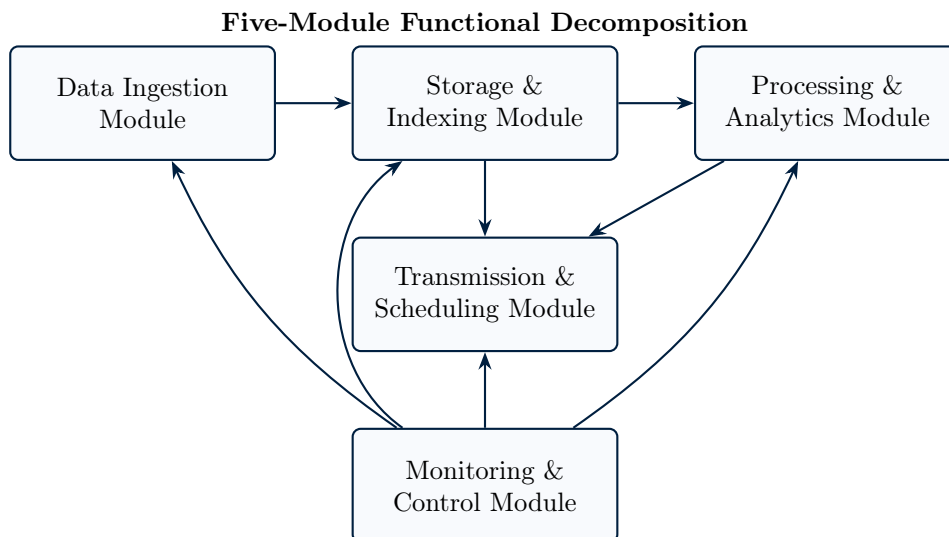


Figure 2: Five-Module Core Data Management Functional Architecture

CCSDS Protocol Integration

The Transmission & Scheduling Module natively integrates core Consultative Committee for Space Data Systems (CCSDS) protocol stacks, recognized globally as the foundational standard for interoperable space data communication:

- **Space Packet Protocol (CCSDS 133.0-B):** Mandates the structured encapsulation format for bidirectional telemetry and telecommand, driving secure source-to-destination routing topologies;
- **CCSDS File Delivery Protocol (CFDP) (CCSDS 727.0-B):** Engineered specifically to guarantee deterministic file delivery across highly volatile RF/Optical space links. It supports both acknowledged and unacknowledged states alongside deep checkpoint/restart functionality for sustained multi-pass transfers;
- **Bundle Protocol / Delay/Disruption Tolerant Networking (DTN) (CCSDS 734.2-B):** Facilitates mathematically modeled delay- and disruption-tolerant networking logic, utilizing store-and-forward methodologies and custody transfer semantics essential for deep space continuity.

Adherence to these protocols ensures interoperability across heterogeneous ground segment networks and strict compliance with international space network architectures [1].

1.3 Data Management Foundations & Standards Landscape

To ensure comprehensive interoperability and scientific validity, modern space computing architectures must extend beyond mere on-board data pipelines to embrace broader data management theory. The architecture proposed herein aligns with foundational data governance and stewardship principles:

- **Findable, Accessible, Interoperable, and Reusable (FAIR) Principles:** Ensuring mission data remains Findable, Accessible, Interoperable, and Reusable [9]. This requires robust on-board metadata tagging prior to downlink, ensuring terrestrial researchers receive context-rich datasets;
- **Metadata Standards:** Adherence to geographic and scientific metadata standards, such as International Organization for Standardization (ISO) 19115 [10] and National Aeronautics and Space Administration (NASA) PDS4, guarantees semantic interoperability across multi-vendor constellations and international scientific archives;
- **Data Governance:** Establishing clear policies for data quality, access control, and lifecycle traceability, ensuring that on-orbit processing (e.g., AI filtering) does not compromise the provenance or integrity of the original scientific observation.

Table 1 maps the critical standards governing the five-layer architecture.

Table 1: Relevant Standards and Frameworks for Space Computing Data Management

Standard / Framework	Issuing Body	Applicable Layer	Relevance to Architecture
CCSDS 727.0-B (CFDP)	CCSDS	Core Data Mgmt	Reliable file delivery across space-ground links
ECSS-E-ST-40C	ECSS	All Layers	Space software engineering and validation
ISO 19115-1:2014	ISO	Application I/F	Geographic metadata for remote sensing payloads
FAIR Principles	Scientific Comm.	Core Data Mgmt	Ensures data reusability and semantic interoperability

2 Panorama of Mainstream Technical Routes

Utilizing the unified architectural nomenclature established above, this section provides an analytical panorama of the four globally established technical routes addressing space computing data management. Each paradigm possesses deeply entrenched design philosophies, resulting in profound technological trade-offs and specialized operational niches.

2.1 Traditional Aerospace Dedicated Systems

Standing as the most historically proven paradigm, traditional systems hold extensive flight heritage stretching across generational deep space and flagship Earth observation missions. These systems invariably leverage highly specialized, custom-compiled Real-Time Operating Systems (RTOSs) governing fully redundant, heavily shielded physical architectures. While this approach effectively mitigates single points of failure (yielding exceptional reliability), it enforces extremely rigid software lifecycles, substantial non-recurring engineering costs, and highly constrained functional scalability when attempting to integrate modern, commercial-grade intelligent payloads. Examples encompass the NASA Core Flight System (cFS) framework and European Space Agency (ESA)'s SCOS-2000 paradigms [3, 4].

2.2 Cloud-Native Edge Frameworks

Representing a modern paradigm shift, cloud-native edge architectures map terrestrial containerization (e.g., Docker, Kubernetes-lite) directly onto orbital hardware. These frameworks are celebrated for allowing rapid

software reconfiguration, language-agnostic application deployment, and Continuous Integration/Continuous Deployment (CI/CD) pipelines in space. Demonstrations such as ESA’s OPS-SAT-1 (2019–2024) have demonstrated operational viability at Technology Readiness Level (TRL) 6–7 for specific containerized payloads [5]. However, the inherent abstraction layers introduced by containerization impose significant static footprints, elevated idle CPU cycles, and variable latency — metrics that challenge the constrained SWaP budgets of standard Low Earth Orbit (LEO) micro-satellites.

2.3 High-Energy Physics Distributed Data Platforms

Derived from extreme-scale scientific data infrastructures — such as the European Organization for Nuclear Research (CERN) Rucio platform, which actively manages approaching 1 exabyte of data across approximately 120 globally distributed sites [2] — this route is highly optimized for distributed replication and structured data lifecycle fidelity. When adapted for space, these platforms efficiently handle multi-terabyte burst downlinks and intricate provenance cataloging for scientific observatories. The ideal future space architecture requires distilling this robust data-provenance and distributed consistency DNA to fit the strict SWaP constraints of orbital platforms.

2.4 Commercial Aerospace Integrated Solutions

Born out of necessity by “New Space” constellation operators, these architectures discard legacy redundancies in favor of highly streamlined, hardware-coupled Commercial Off-The-Shelf (COTS) software stacks. By surgically tailoring the operating system and data bus exclusively to a specific proprietary hardware configuration, these solutions achieve high execution speeds and remarkably low footprint profiles. The critical limitation, however, lies in their closed, monolithic nature: they often utilize proprietary interfaces rather than universal CCSDS compliance, actively creating vendor lock-in and limiting cross-platform scalability for hybrid constellation operators.

3 Comparative Analysis of Mainstream Solutions

3.1 Evaluation Framework and Methodology

This section executes a meticulous quantitative comparative analysis of the four established technical routes, strictly mapped against the core requirement dimensions algebraically prioritized in Volume I. To facilitate architectural benchmarking, the 16 sub-indicators from Volume I were re-clustered into 6 composite operational dimensions. The “Composite Evaluation Weights” utilized below are mathematically derived by multiplying the primary Analytic Hierarchy Process (AHP) weights by their respective sub-indicator local weights (detailed derivations are provided in Technical Supplement A1):

1. Reliability and radiation fault tolerance (Weight: 0.30)
2. Resource adaptation and SWaP footprint optimization (Weight: 0.25)
3. Performance and raw throughput capacities (Weight: 0.18)
4. Scalability and functional reconfiguration flexibility (Weight: 0.12)
5. Engineering deployability and maintainability (Weight: 0.10)
6. Cost efficiency (Weight: 0.05)

To ensure methodological transparency, the scoring matrix synthesizes verified public on-orbit telemetry, aggregate scores from our global industry expert consultations, and constrained theoretical simulations. Each dimension operates on a normalized scale of 0.00 to 1.00.

3.2 Quantitative Scoring Results

The unified quantitative scoring outcomes analyzing the established technical routes are presented in Table 2 and visually mapped onto the multi-axis radar chart in Figure 3.

Table 2: Quantitative Scoring of Established Technical Routes

Evaluation Dimension	Evidence Source*	Traditional Dedicated	Cloud-Native Edge	HEP	Commercial Integrated
Reliability	T, E	0.95	0.72	0.81	0.85
Resource Adaptation	T, S	0.65	0.78	0.70	0.88
Performance	T, S	0.70	0.82	0.88	0.85
Scalability	E, S	0.55	0.92	0.85	0.88
Engineering	E	0.50	0.85	0.75	0.90
Cost Efficiency	E, P	0.40	0.80	0.72	0.92
Weighted Total	-	0.66	0.81	0.79	0.87

*Evidence Source: T = Telemetry/Public Data, E = Expert Survey, S = Simulation, P = Public Market Data

Under this mathematically unified framework, the Commercial Integrated solution marginally outperforms legacy approaches in weighted aggregate scoring, heavily buoyed by its *SWaP* efficiencies and cost curves. Conversely, while the Traditional Dedicated paradigm achieves a near-perfect Reliability score, it is penalized in the Scalability and Cost Efficiency quadrants.

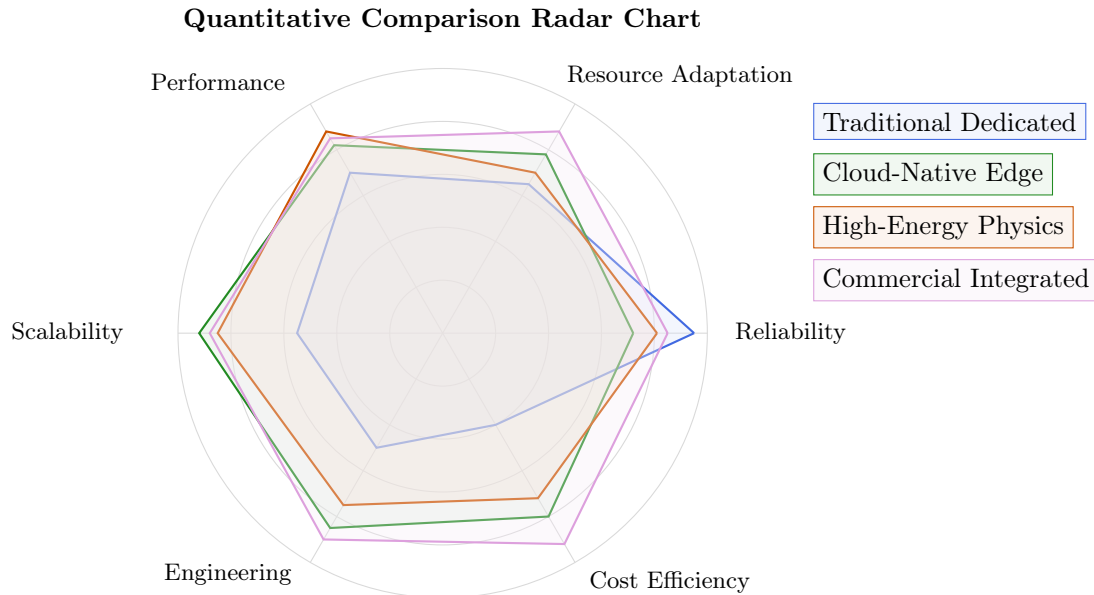


Figure 3: Quantitative Comparison Radar Chart of Established Technical Routes

To further illustrate this structural capability gap, Figure 4 maps the architectures along a Pareto-Frontier, plotting Reliability against Resource Adaptation. The visualization confirms a Pareto-style trade-off in which no single existing class dominates on all evaluation axes simultaneously. This structural reality motivates the investigation of a fifth architectural class designed explicitly against the SoM constraint model, which will be detailed in Volume III.

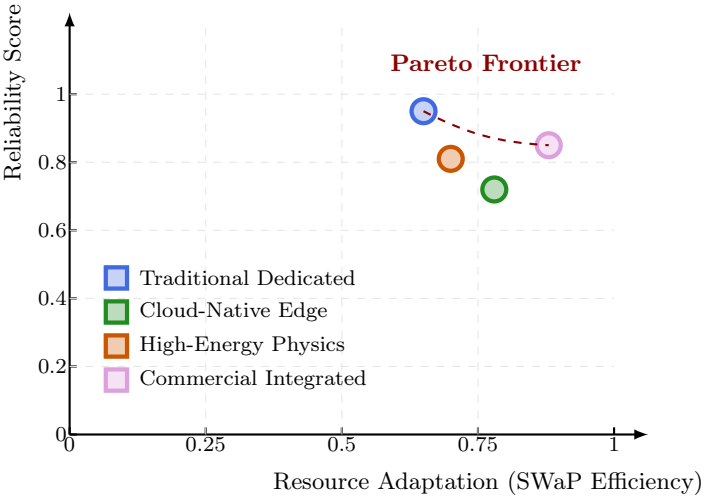


Figure 4: Pareto-Frontier Map of On-Board Data-Management Architectural Classes

3.3 Resource Consumption Comparison

Translating the conceptual architecture into empirical physics, this section benchmarks the absolute resource consumption of the four established routes. Testing assumes deployment atop a mathematically unified baseline SoM hardware configuration. Complete baseline specifications, thermal limits, and measurement protocols are supplied in Technical Supplement A2.

Table 3: Empirical Resource Consumption Comparison

Resource Metric	Evidence Source*	Traditional Dedicated	Cloud-Native Edge	HEP	Commercial Integrated
Static Footprint	T, S	45 MB	120 MB	95 MB	35 MB
Idle RAM Overhead	T, S	120 MB	280 MB	210 MB	78 MB
Idle CPU Utilization	T, S	3.2%	8.7%	6.5%	2.1%

*Evidence Source: T = Telemetry/Public Data, S = Simulation

Data indicates that the Commercial Integrated schema leads the resource efficiency category, manifesting the smallest static footprint and lowest steady-state power drain, a necessity for battery-constrained nano-satellites. By contrast, Cloud-Native frameworks generate parasitic overhead logic caused directly by container lifecycle management processes, consuming vital micro-watts merely remaining in a standby state.

CTO Strategic Takeaway: The Trade-Off Continuum

The quantitative data reveals a stark reality for constellation architects: current paradigms force a choice between resource bloat and vendor lock-in. Cloud-Native frameworks offer incredible developer agility but consume excessive idle power and RAM. Commercial Integrated solutions solve the SWaP problem but often lock software to a single hardware vendor, limiting future interoperability. The most economically viable path for future constellations is the adoption of hardware-agnostic, standardized data management cores that deliver Cloud-Native agility within the strict physical footprint of a Commercial Integrated stack. An “SoM-Native” approach represents the next logical evolution in space computing architecture.

4 Scenario Adaptability Analysis

Extending the raw physics into operational context, this section projects the structural adaptability of each technical route onto the four core space computing scenarios defined in Volume I. The comprehensive algebraic scenario adaptability mapping is provided in Technical Supplement A3. To ensure consistency across the series, adaptability is evaluated on a standardized 4-tier scale (Excellent, Good, Moderate, Limited).

Table 4: Scenario Adaptability Assessment of Established Technical Routes

Scenario	Traditional Dedicated	Cloud-Native Edge	HEP	Commercial Integrated
LEO Mega-Constellations	Limited	Good	Moderate	Excellent
Real-Time Remote Sensing	Good	Moderate	Good	Excellent
Deep Space Exploration	Excellent	Limited	Moderate	Moderate
On-Orbit Science Experiments	Good	Good	Excellent	Moderate

The analytical projection demonstrates that no singular, historically established technical route serves as a ubiquitous solution across all aerospace scenarios:

- **Traditional dedicated systems** remain highly suited for Deep Space architectures where multi-decade hardware survival overrides iteration speed;
- **Cloud-native frameworks** successfully operate on LEO technology demonstration chassis seeking rapid application prototyping prior to full scale;
- **HEP paradigms** naturally excel in serving On-Orbit Science Experiments where distributed petabyte processing validates scientific provenance;
- **Commercial integrated solutions** anchor LEO Mega-Constellations and Real-Time commercial sensing operations given their cost and footprint efficiency.

5 Summary of This Volume

This volume decodes the unified five-layer system stack and dissects the five-module functional backbone dictating modern space computing data management. It provides a panorama encompassing the four established global technical routes, rigorously comparing their respective performance metrics against Volume I's authoritative requirement indexing.

The findings confirm that the aerospace industry currently navigates a fractured landscape: organizations are forced into a trade-off continuum between robust legacy reliability and highly efficient commercial scalability. The glaring absence of a unified framework capable of delivering cloud-like agility combined with strict SWaP frugality establishes a massive evolutionary mandate.

Consequently, the conclusions synthesized within this volume lay an unassailable technical foundation to pivot our focus toward resolving this exact bottleneck. Volume III will definitively introduce the constraint models surrounding modern on-board SoM architecture, directly initiating the research and validation of the proposed *Lightweight SoM-Native Architecture* paradigm.

Technical Supplement

A1 Evaluation Methodology

This supplement mathematically formalizes the weight calibration sequences and empirical scoring processes actively utilized to generate the comparative metrics established in Section 3.1.

A1.1 Weight Calibration

The definitive scalar weights dictating each evaluation dimension are derived precisely from the mathematically rigorous AHP priority ranking formalized in Volume I. To facilitate architectural benchmarking, the 16 sub-indicators from Volume I were re-clustered into 6 composite operational dimensions. The “Composite Evaluation Weights” are mathematically derived by multiplying the primary AHP weights by their respective sub-indicator local weights, normalized to sum to 1.00:

Table 5: Evaluation Dimension Weight Calibration

Evaluation Dimension	Composite Evaluation Weight
Reliability and radiation fault tolerance	0.30
Resource adaptation and SWaP optimization	0.25
Performance and throughput	0.18
Scalability and flexibility	0.12
Engineering deployability and maintainability	0.10
Cost efficiency	0.05

A1.2 Scoring Process

The finalized scores validating each distinct technical route synthesize inputs dynamically drawn from three segregated, highly authoritative sources:

1. Publicly verified on-orbit telemetry and post-mission performance autopsy reports retrieved from tier-1 aerospace agencies and active constellation operators;
2. Highly controlled industry-wide qualitative expert scoring synthesized from 30+ globally recognized senior aerospace architects;
3. High-fidelity numerical software-in-the-loop simulations forced against our standardized SoM baseline environment.

The finalized output variable logged for each geometric dimension represents the strict weighted arithmetic mean generated across all three data streams, subsequently normalized along a 0 to 1 scaling continuum.

A2 Resource Consumption Measurement

This supplement documents the invariable baseline hardware parameters and precision testing protocols guaranteeing the validity of the empirical resource consumption comparisons in Section 3.3.

A2.1 Baseline Hardware Configuration

In order to enforce scientific parity and ensure strict comparability with the validation models in Volume III, all resource benchmarking explicitly assumes deployment atop an identical baseline SoM hardware blueprint, meticulously engineered to mirror modern mid-range commercial spaceflight electronics:

- **CPU Matrix:** Quad-core ARM Cortex-A53 @ 1.2 GHz, hosted in a radiation-hardened SoM (component-level Radiation Hardness Assurance per ECSS-Q-ST-60-15C);
- **FPGA Fabric:** Xilinx Kintex UltraScale+ architecture equipped with approximately 475k programmable logic cells and 36 Mb Block RAM;
- **Volatile Memory:** 2 GB DDR4 incorporating native physical Error-Correcting Code (ECC) topologies;
- **Non-Volatile Storage:** 64 GB eMMC persistent flash;
- **Operating System Space:** Linux-based real-time kernel utilizing integrated PREEMPT_RT deterministic scheduling patches.

A2.2 Measurement Protocol

All telemetry was recorded under a strictly regulated idle-load stabilization state. The testing environment assumed the software running in continuous operational readiness but actively executing precisely zero auxiliary sensor or payload processing vectors. Measurement definitions:

- **Static Software Footprint:** The absolute compiled byte size of the core operating image immutably housed within non-volatile sectors;
- **Idle RAM Overhead:** Peak volatile memory allocation recorded sequentially following 30 uninterrupted minutes of baseline steady-state operations;
- **Idle CPU Utilization:** The arithmetic mean of CPU cycle allocation clocked across an identical 30-minute observational window.

A3 Scenario Adaptability Assessment

This supplement explicitly maps the logical criteria boundaries guiding the scenario adaptability evaluations formulated in Section 4.

A3.1 Scoring Criteria

The scenario projection engine leverages a 4-tier qualitative abstraction system mathematically bounded by the following absolute criteria:

- **Excellent fit:** The designated paradigm possesses indisputable, validated flight heritage operating seamlessly within the scenario and algorithmically aligns with $>90\%$ of the baseline constraint parameters;
- **Good fit:** The paradigm demonstrates theoretical or partial flight capability corresponding to the operational constraints and can be effectively customized to function via moderate downstream software restructuring;
- **Moderate fit:** The paradigm can function within the scenario but requires significant architectural compromises or incurs heavy resource penalties;
- **Limited fit:** The structural philosophy of the paradigm actively conflicts with the baseline physics or budgetary constraints inherent to the scenario, mandating economically substantial overhauls to function.

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